

Audit Instantons: Non-Perturbative Topological Defects in Expert Auditing

Instanton

A Mathematical Formalization Inspired by String Theory Worldsheet Instantons

Instanton

SCX

Abstract

We introduce the concept of **audit instantons** — non-perturbative, topologically protected failures of expert consensus in the SCX (Situs Consensus eXpert) auditing framework. Drawing structural inspiration from worldsheet instantons in string theory, we demonstrate that pointwise Yajie consensus auditing is analogous to perturbative expansion: it systematically misses globally correlated expert failures concentrated in low-data-density regions of the Situs manifold.

We provide a rigorous mathematical formulation: the Situs manifold $\mathcal{M}$ equipped with a data density function $\rho(x)$; low-density sublevel sets $\mathcal{M}_\varepsilon = \{x \in \mathcal{M} : \rho(x) < \varepsilon\}$; and an *audit instanton* defined as a non-contractible 1-cycle $\gamma \subset \mathcal{M}$ on which all experts deviate consistently in the same direction — locally exact ($d_1 A = 0$ on γ) but globally non-trivial ($\gamma \notin \text{im}(d_0)$). Detection proceeds via persistent homology of the density filtration, where long-lived 1-cycles in the persistence diagram correspond to audit instanton candidates.

We prove the main structural theorem: every non-trivial 1-homology class of the Situs manifold contains at least one audit instanton candidate, provided the expert output field satisfies mild regularity conditions. We further show that audit instantons are invisible to the standard Yajie consensus score — precisely the “non-perturbative” character that makes them dangerous. A concrete detection protocol is given.

*we introduce***Instanton***audit instantons*—*SCX**Situs**Instanton**Proof**Yajie**Missing**Situs*
 $\rho(x)$ *Situs* $\mathcal{M}$ $\mathcal{M}_\varepsilon = \{x \in \mathcal{M} : \rho(x) < \varepsilon\}$ **Instanton**—*define* $\mathcal{M}$ *1-* γ *all*— $d_1 A =$
 0 $\gamma \gamma \notin \text{im}(d_0)$ *where* *1-Instanton*
*Proof**theorems**satisfying**Situs**each**1-Instanton**Proof**Instanton**Yajie**Score*—“”

Keywords: audit instantons, persistent homology, discrete Hodge theory, topological data analysis, expert auditing, non-perturbative effects, SCX framework
Keywords InstantonHodgeSCX

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1 Introduction

1.1 Motivation from Physics

In quantum field theory and string theory, *instantons* are non-perturbative field configurations: classical solutions to the Euclidean equations of motion with finite action that contribute terms of order $\sim \exp(-1/g^2)$ to the path integral, where g is the coupling constant [4, 5]. Crucially, such contributions are identically zero in any finite-order Taylor expansion around $g = 0$ — they are *invisible to perturbation theory*.

Instanton $\sim \exp(-1/g^2)$ where $g \rightarrow 0$ —

In string theory specifically, **worldsheet instantons** arise from string worldsheets wrapping holomorphic curves in the compactification manifold. Their contribution is $\sim \exp(-A/\alpha')$, where A is the area of the wrapped curve and α' is the string tension. These effects are:

- (i) **Non-perturbative**: invisible to the genus expansion (Feynman diagrams);
- (ii) **Topological**: classified by the homotopy/homology of the target space;
- (iii) **Exponentially suppressed** in the small- g regime, yet dominant in certain corners of moduli space.

in particular *Instanton* $\sim \exp(-A/\alpha')$ where A/α' (i) Feynman (ii) / (iii) g

1.2 The Auditing Analogy

The SCX (Situs Consensus eXpert) framework [1] performs expert auditing on a data manifold $\mathcal{M}$ using discrete Hodge theory and gauge-theoretic consensus measures. The standard Yajie audit operates *pointwise*: at each data point $x \in \mathcal{M}$, it compares expert outputs and assigns a consensus score based on local agreement. This is structurally analogous to *perturbative expansion* in quantum field theory — each data point is treated independently, and the audit result is the sum of local contributions.

SCX Situs Hodge $\mathcal{M}$ Yajie $\mathcal{M}$ each x Consistency Score — each

The central insight of this paper is that **pointwise audit is perturbative audit**, and it suffers from the same fundamental blindness: it cannot detect *globally correlated, topologically protected* failure modes that we term **audit instantons**. These are configurations where:

- (1) The data density $\rho(x)$ is low (the “weak-coupling” regime of audit);
- (2) All experts deviate in the *same* direction — producing *spurious consensus* that looks like high agreement to pointwise Yajie;
- (3) The deviation field is locally consistent (closed form) but globally non-trivial (not exact) — a topological obstruction invisible to local analysis.

Core i.e. *Instantons* satisfying (1) $\rho(x)$ “(2) all — Yajie Consistency (3) —

1.3 Our Contributions

This paper provides the first rigorous mathematical formalization of audit instantons. Specifically, we:

- (C1) **Formal Definition** (Section 3): Define the Situs manifold $(\mathcal{M}, \rho, \mathbf{A})$ with data density ρ and audit vector field $\mathbf{A}$, and give a cohomological definition of audit instantons as non-contractible 1-cycles with locally exact but globally non-trivial deviation fields.

- (C2) **Detection Theory** (Section 4): Show that persistent homology of the density sublevel filtration provides a systematic detection mechanism: 1-cycles with large persistence (long lifetime in the barcode) are audit instanton candidates.
- (C3) **Structural Theorem** (Theorem 4.4): Prove that every non-trivial 1-homology class in $\mathcal{M}$ contains at least one audit instanton candidate, under mild conditions on the expert output field.
- (C4) **Invisibility to Yajie** (Section 5): Prove that the standard Yajie consensus score evaluated on an audit instanton cycle systematically overestimates audit quality — the very definition of a non-perturbative effect.
- (C5) **Detection Protocol** (Section 6): Provide a concrete, algorithmically implementable protocol for detecting audit instantons in real SCX deployments.

This paper provides the first **Instanton** *(C1) define* ρ **Situs** $(\mathcal{M}, \rho, \mathbf{A})$ *Instanton* *define* (C2) *4Proof* — *1-Instanton* (C3) *theorem* *theorem* 4.4 *Proof* $\mathcal{M}$ *each* *1-Instanton* (C4) *Yajie* *5Proof* *Yajie Score* *Instanton* *6Instanton*

2 Mathematical Preliminaries

2.1 The Situs Manifold

Situs

Definition 2.1 (Situs Manifold / Situs). Let $\mathcal{M}$ be a compact, connected, smooth Riemannian n -manifold with boundary $\partial\mathcal{M}$ (possibly empty). The **Situs manifold** is the triple $(\mathcal{M}, \rho, \mathbf{A})$, where:

- (i) $\rho : \mathcal{M} \rightarrow \mathbb{R}_{\geq 0}$ is a smooth (or at least C^2) **data density function**, normalized so that $\int_{\mathcal{M}} \rho(x) d\text{vol}_{\mathcal{M}}(x) = 1$. In practice, ρ is estimated from the empirical distribution of training or audit data points via kernel density estimation.
- (ii) $\mathbf{A} = \{E_1, E_2, \dots, E_M\}$ is a finite set of $M \geq 2$ **experts**, where each $E_i : \mathcal{M} \rightarrow \mathbb{R}^d$ is a vector-valued function representing the expert's output (prediction, classification, or scalar field) at each point $x \in \mathcal{M}$.

The **dimension** n of $\mathcal{M}$ is typically the intrinsic dimension of the data representation space (e.g., after dimensionality reduction).

define 2.1 **Situs** $\mathcal{M}$ n **Riemann** $\partial\mathcal{M}$ **Situs** $(\mathcal{M}, \rho, \mathbf{A})$ *where* (i) $\rho : \mathcal{M} \rightarrow \mathbb{R}_{\geq 0}$ C^2 *such that* $\int_{\mathcal{M}} \rho(x) d\text{vol}_{\mathcal{M}}(x) = 1$ ρ (ii) $\mathbf{A} = \{E_1, \dots, E_M\}$ $M \geq 2$ *each* $E_i : \mathcal{M} \rightarrow \mathbb{R}^d$ *each* $x \in \mathcal{M}$

Remark 2.2 (Discrete vs. Continuous /). In SCX practice, $\mathcal{M}$ is discretized as a simplicial complex K (or graph/cell complex) with vertices corresponding to data points. All definitions in this paper extend naturally to the discrete setting via discrete Hodge theory [2]. We work in the continuous category for clarity, but all constructions have well-defined discrete analogues.

remark **SCX** $\mathcal{M}$ K */all* *define* *Hodge* *all* *define*

2.2 Data Density and Sublevel Sets

Definition 2.3 (Density Sublevel Sets /). For any threshold $\varepsilon \geq 0$, the ε -**sublevel set** of the data density is:

$$\mathcal{M}_\varepsilon = \{x \in \mathcal{M} : \rho(x) \leq \varepsilon\}. \quad (1)$$

By convention, $\mathcal{M}_0 = \{x : \rho(x) = 0\}$ and $\mathcal{M}_\infty = \mathcal{M}$.

The family $\{\mathcal{M}_\varepsilon\}_{\varepsilon \geq 0}$ forms a **filtration** of $\mathcal{M}$: $\mathcal{M}_0 \subseteq \mathcal{M}_{\varepsilon_1} \subseteq \mathcal{M}_{\varepsilon_2} \subseteq \mathcal{M}$ for $0 \leq \varepsilon_1 \leq \varepsilon_2$.

The **critical density** ρ_{crit} is a distinguished threshold below which expert behavior is deemed unreliable due to data sparsity. Formally, ρ_{crit} may be defined as:

$$\rho_{\text{crit}} = \inf\{\varepsilon > 0 : \text{expert predictions on } \mathcal{M}_\varepsilon \text{ are statistically indistinguishable from their behavior on } \mathcal{M} \setminus \mathcal{M}_\varepsilon\} \quad (2)$$

define 2.3 any $\varepsilon \geq 0$ - $\mathcal{M}_\varepsilon = \{x \in \mathcal{M} : \rho(x) \leq \varepsilon\}$ $\{\mathcal{M}_\varepsilon\}_{\varepsilon \geq 0}$ $\mathcal{M}$ ρ_{crit} define $\mathcal{M}_\varepsilon$ $\mathcal{M} \setminus \mathcal{M}_\varepsilon$

2.3 Discrete Hodge Theory (Recap)

Hodge

We briefly recall the essential elements of discrete Hodge theory on $\mathcal{M}$, as used in SCX [2]. Let $\Omega^k(\mathcal{M})$ denote the space of smooth differential k -forms on $\mathcal{M}$.

Definition 2.4 (Cochain Complex /). The **de Rham complex** on $\mathcal{M}$ is:

$$0 \longrightarrow \Omega^0(\mathcal{M}) \xrightarrow{d_0} \Omega^1(\mathcal{M}) \xrightarrow{d_1} \Omega^2(\mathcal{M}) \xrightarrow{d_2} \dots \xrightarrow{d_{n-1}} \Omega^n(\mathcal{M}) \longrightarrow 0, \quad (3)$$

where each $d_k : \Omega^k(\mathcal{M}) \rightarrow \Omega^{k+1}(\mathcal{M})$ is the exterior derivative, satisfying $d_{k+1} \circ d_k = 0$.

The k -**th de Rham cohomology** is $H_{\text{dR}}^k(\mathcal{M}) = \ker(d_k) / \text{im}(d_{k-1})$.

The **Hodge Laplacian** on k -forms is:

$$\Delta_k = d_{k-1} \delta_k + \delta_{k+1} d_k, \quad (4)$$

where $\delta_k = (-1)^{n(k-1)+1} \star d_{n-k} \star$ is the codifferential and $\star$ is the Hodge star operator induced by the Riemannian metric.

define $\mathcal{M}$ **de Rham** (3) where d_k satisfying $d_{k+1} \circ d_k = 0$ k **de Rham** $H_{\text{dR}}^k(\mathcal{M}) = \ker(d_k) / \text{im}(d_{k-1})$ **Hodge Laplacian** (4) where $\delta_k \star$ Hodge

Definition 2.5 (Expert Output 1-Form / 1-). Given M experts and a reference consensus direction (e.g., the Yajie center or a ground-truth label), define for each expert E_i the **deviation 0-form** $f_i \in \Omega^0(\mathcal{M})$:

$$f_i(x) = \|E_i(x) - \bar{E}(x)\|_{\text{signed}}, \quad (5)$$

where $\bar{E}(x)$ is the consensus output at x and $\|\cdot\|_{\text{signed}}$ is a signed distance (positive for deviation in one direction, negative for the opposite). The **audit 1-form** is then:

$$A = d_0 \left(\frac{1}{M} \sum_{i=1}^M f_i \right) \in \Omega^1(\mathcal{M}), \quad (6)$$

representing the spatial gradient of the average expert deviation. In the discrete setting, A is a 1-cochain on the simplicial complex.

define 1- M each E_i **define 0-** $f_i(x) = \|E_i(x) - \bar{E}(x)\|_{\text{signed}}$ where $\bar{E}(x)$ **1-** $A = d_0(M^{-1} \sum_i f_i)$

2.4 Persistent Homology

Definition 2.6 (Persistence Module /). Let $\{\mathcal{M}_\varepsilon\}_{\varepsilon \geq 0}$ be the density sublevel filtration. For each $k \geq 0$, the k -th homology functor $H_k(\cdot; \mathbb{F})$ (with coefficients in a field $\mathbb{F}$, typically $\mathbb{Z}_2$ or $\mathbb{Q}$) applied to the filtration yields a **persistence module** $\mathbb{V}_k = \{H_k(\mathcal{M}_\varepsilon; \mathbb{F})\}_{\varepsilon \geq 0}$ with linear maps $v_{\varepsilon_1, \varepsilon_2} : H_k(\mathcal{M}_{\varepsilon_1}; \mathbb{F}) \rightarrow H_k(\mathcal{M}_{\varepsilon_2}; \mathbb{F})$ induced by inclusions $\mathcal{M}_{\varepsilon_1} \hookrightarrow \mathcal{M}_{\varepsilon_2}$ for $\varepsilon_1 \leq \varepsilon_2$.

define 2.6 $\{\mathcal{M}_\varepsilon\}_{\varepsilon \geq 0}$ each $k \geq 0$ $H_k(\cdot; \mathbb{F})$ $\mathbb{V}_k = \{H_k(\mathcal{M}_\varepsilon; \mathbb{F})\}_{\varepsilon \geq 0}$

By the structure theorem of persistent homology [6, 7], $\mathbb{V}_k$ decomposes into a direct sum of **interval modules**:

$$\mathbb{V}_k \cong \bigoplus_{j \in \mathcal{J}_k} \mathbb{I}_{[b_j, d_j]}, \quad (7)$$

where each $\mathbb{I}_{[b, d]}$ is an indecomposable module supported on the interval $[b, d]$, with **birth** b_j (the density threshold at which the homology class appears) and **death** d_j (the threshold at which it merges with another class or becomes trivial).

theorem $\mathbb{V}_k$ each $\mathbb{I}_{[b, d]}$ $[b, d]$ b_j d_j

Definition 2.7 (Persistence Diagram /). The k -dimensional **persistence diagram** is the multiset:

$$\mathcal{D}_k = \{(b_j, d_j) \in \mathbb{R}^2 : b_j < d_j, j \in \mathcal{J}_k\}, \quad (8)$$

together with all points on the diagonal $\{(b, b) : b \geq 0\}$ taken with infinite multiplicity (representing classes of zero persistence).

The **persistence** (or **lifetime**) of a homology class is $([\gamma]) = d_j - b_j$. Classes with large persistence are considered **topologically significant**, while those with small persistence are typically attributed to noise [8].

define k $\mathcal{D}_k = \{(b_j, d_j) : b_j < d_j\}$ all $([\gamma]) = d_j - b_j$

3 Audit Instantons: Formal Definition

Instanton define

3.1 The Audit Vector Field and Its Cohomology

Consider an audit pool of M experts on the Situs manifold $\mathcal{M}$. Let $\{x_\alpha\}_{\alpha=1}^N \subset \mathcal{M}$ be a set of N audit data points (which may or may not coincide with training data). At each point x , expert E_i produces a prediction $E_i(x) \in \mathbb{R}^d$.

Definition 3.1 (Expert Deviation Field /). The **signed expert deviation field** is a function $\mathbf{f} : \mathcal{M} \rightarrow \mathbb{R}^M$ defined componentwise:

$$\mathbf{f}(x) = (f_1(x), f_2(x), \dots, f_M(x)), \quad (9)$$

where each $f_i : \mathcal{M} \rightarrow \mathbb{R}$ is the signed deviation of expert i from a reference value $y(x)$ (e.g., ground truth or Yajie consensus):

$$f_i(x) = \langle E_i(x) - y(x), \mathbf{n}(x) \rangle, \quad (10)$$

with $\mathbf{n}(x)$ a distinguished normal direction field (for scalar outputs, $\mathbf{n}(x) \equiv 1$ and $f_i(x) = E_i(x) - y(x)$).

define 3.1 $\mathbf{f} : \mathcal{M} \rightarrow \mathbb{R}^M$ where each $f_i(x) = \langle E_i(x) - y(x), \mathbf{n}(x) \rangle$ i $y(x)$

Definition 3.2 (Audit 1-Form / 1-). The **audit 1-form** $A \in \Omega^1(\mathcal{M})$ is the exterior derivative of the *mean* signed deviation:

$$A = d_0 \bar{f}, \quad \text{where} \quad \bar{f}(x) = \frac{1}{M} \sum_{i=1}^M f_i(x). \quad (11)$$

Equivalently, in local coordinates $(x^1, \dots, x^n)$:

$$A = \sum_{\mu=1}^n \frac{\partial \bar{f}}{\partial x^\mu} dx^\mu. \quad (12)$$

define 3.21-1- $A \in \Omega^1(\mathcal{M})$ $A = d_0 \bar{f}$

Proposition 3.3 (Automatic Closedness /). By construction, $d_1 A = d_1 d_0 \bar{f} = 0$. Hence A is always a closed 1-form: $A \in \ker(d_1) = Z^1(\mathcal{M})$.

Proposition 3.3d1 $A = d_1 d_0 \bar{f} = 0$ therefore $A \in Z^1(\mathcal{M})$

Proposition 3.4 (Exactness Condition /). The audit 1-form A is exact (i.e., $A \in \text{im}(d_0) = B^1(\mathcal{M})$) if and only if the mean deviation $\bar{f}$ is well-defined as a global smooth function. In that case, A lies in the trivial cohomology class: $[A] = 0 \in H_{dR}^1(\mathcal{M})$.

If $\bar{f}$ has monodromy — i.e., its integral around a closed loop γ is non-zero:

$$\oint_{\gamma} A = \oint_{\gamma} d_0 \bar{f} \neq 0, \quad (13)$$

then $\bar{f}$ is not globally single-valued, and A represents a non-trivial cohomology class $[A] \neq 0 \in H_{dR}^1(\mathcal{M})$.

Proposition 3.41- A i.e. $A \in B^1(\mathcal{M})$ if and only if $\bar{f}$ define $A \bar{f} \longrightarrow \gamma \longrightarrow \bar{f} A$

3.2 Definition of Audit Instanton

Instanton define

Definition 3.5 (Audit Instanton / Instanton). Let $(\mathcal{M}, \rho,)$ be a Situs manifold and $A \in \Omega^1(\mathcal{M})$ the audit 1-form. An **audit instanton** is a piecewise-smooth closed curve $\gamma : S^1 \rightarrow \mathcal{M}$ (a 1-cycle) satisfying the following three conditions:

(AI1) Low-Density Localization / : γ is contained in a low-density region:

$$\gamma \subset \mathcal{M}_{\rho_{\text{crit}}} = \{x \in \mathcal{M} : \rho(x) < \rho_{\text{crit}}\}. \quad (14)$$

(AI2) Local Consistency / Consistency: The audit 1-form is locally exact along γ :

$$\forall p \in \gamma, \exists \text{ a neighborhood } U_p \subset \mathcal{M} \text{ and } g_p \in \Omega^0(U_p) \text{ such that } A|_{U_p} = d_0 g_p. \quad (15)$$

This holds automatically by Proposition 3.4 — the substantive content is that the local primitive g_p must have *bounded variation* along γ : $\|\nabla g_p\|_{L^\infty(\gamma \cap U_p)} \leq C$ for some universal constant C .

(AI3) Global Non-Triviality / : The cycle is not a boundary in the low-density region:

$$\gamma \notin \text{im}(\partial_2) \text{ in } H_1(\mathcal{M}_{\rho_{\text{crit}}}; \mathbb{Z}), \quad (16)$$

where $\partial_2 : C_2(\mathcal{M}_{\rho_{\text{crit}}}) \rightarrow C_1(\mathcal{M}_{\rho_{\text{crit}}})$ is the simplicial boundary operator. Equivalently, γ represents a *non-trivial* homology class: $[\gamma] \neq 0 \in H_1(\mathcal{M}_{\rho_{\text{crit}}}; \mathbb{Z})$.

Moreover, the **holonomy** (circulation) of A around γ is non-zero:

$$\Phi(\gamma) = \oint_{\gamma} A > \eta, \quad (17)$$

where $\eta > 0$ is a significance threshold (e.g., the expected noise level of the audit 1-form).

define 3.5 Instanton $(\mathcal{M}, \rho, \gamma)$ *Situs A 1-Instanton* satisfying $\gamma : S^1 \rightarrow \mathcal{M}$ 1-**(AI1)** $\gamma \subset \mathcal{M}_{\rho_{\text{crit}}}$ **(AI2) Consistency** $\gamma \gamma$ **(AI3)** $\gamma \notin \text{im}(\partial_2) H_1(\mathcal{M}_{\rho_{\text{crit}}}; \mathbb{Z})$ $A \gamma \Phi(\gamma) = \oint_{\gamma} A > \eta$

Remark 3.6 (Physical Interpretation /). The three conditions capture the essential “instanton-ness” of the configuration:

- (AI1) ensures we are in the “weak-coupling” (low-data) regime where non-perturbative effects dominate — analogous to instantons contributing at small g in QFT.
- (AI2) ensures that *locally* the deviation field looks like a gradient of a well-behaved function — analogous to the classical instanton solving the equations of motion. Pointwise Yajie audit sees local consistency and concludes “high consensus.”
- (AI3) ensures that *globally* the deviation accumulates non-zero net error — the topological obstruction that pointwise audit cannot detect. This is the “instantonic charge” or “winding number.”

remark “Instanton” (AI1) “(AI2) — Yajie Consistency Conclusion” (AI3) — i.e. “Instanton”

3.3 Non-Perturbative Character

The defining property of audit instantons — and what makes them potentially catastrophic — is their **invisibility to pointwise audit**. We formalize this as follows.

Proposition 3.7 (Pointwise Audit Blindness /). *Let γ be an audit instanton. The standard pointwise Yajie audit score $\text{Yajie}(x)$, defined as a local function of expert outputs at x alone, satisfies:*

$$\inf_{x \in \gamma} \text{Yajie}(x) \geq \text{Yajie}_{\text{threshold}}, \quad (18)$$

while the ground-truth error integrated along γ exceeds the audit tolerance:

$$\frac{1}{|\gamma|} \oint_{\gamma} |\bar{f}(x)| dx \gg \varepsilon_{\text{tol}}. \quad (19)$$

Proposition 3.7 γ Instanton Yajie Score $\text{Yajie}(x)$ satisfying $\inf_{x \in \gamma} \text{Yajie}(x) \geq \text{Yajie}_{\text{threshold}}$

Proof Sketch / Proof Sketch. Condition (AI2) guarantees that at each point $x \in \gamma$, the local deviation has bounded variation — all experts deviate “smoothly” in the same direction. Pointwise Yajie, which only compares expert outputs at x , sees this as consensus (everyone agrees) and assigns a high score. However, condition (AI3) ensures that the accumulated deviation around the full cycle γ is non-zero — the consensus is *spurious* and the total error is systematic. This is a direct analog of the perturbative expansion’s inability to detect $\sim \exp(-1/g^2)$ terms. $\square$

Proof Sketch(AI2) γ x — all “Yajie x all however (AI3) γ — $\sim \exp(-1/g^2)$

4 Detection Theory via Persistent Homology

4.1 Persistent Homology of the Density Filtration

The density sublevel filtration $\{\mathcal{M}_\varepsilon\}_{\varepsilon \geq 0}$ captures the topology of data-sparse regions at multiple scales. As ε increases, new topological features (connected components, cycles, voids) are born, persist for some ε -range, and eventually die.

Definition 4.1 (Density Filtration Barcode /). The **1-dimensional barcode** of the density filtration is the set of intervals:

$$\mathcal{B}_1 = \{(b_j, d_j)\}_{j \in \mathcal{J}_1}, \quad (20)$$

where each interval corresponds to a 1-cycle (“hole” or “tunnel”) that:

- Is **born** at density b_j : the 1-cycle first appears as a hole in $\mathcal{M}_{b_j}$;
- **Dies** at density d_j : the hole is filled (becomes a boundary) in $\mathcal{M}_{d_j}$.

Equivalently, this is the persistence diagram $\mathcal{D}_1$ of the superlevel set filtration of ρ , viewed in the reverse parametrization.

define 4.11 $\mathcal{B}_1 = \{(b_j, d_j)\}_{j \in \mathcal{J}_1}$ each 1-“” b_j $\mathcal{M}_{b_j}$ d_j $\mathcal{M}_{d_j}$

The key intuition is that **long-lived 1-cycles** (those with large $= d_j - b_j$) correspond to *robust* low-density regions — tunnels or loops in the data manifold that remain empty of data over a wide range of density thresholds. These are precisely the regions where audit instantons can form.

1- = $d_j - b_j$ — Instanton

4.2 Correspondence: 1-Cycles and Audit Instanton Candidates

1-Instanton

We now establish the connection between persistent 1-cycles and audit instantons.

Definition 4.2 (Audit Instanton Candidate / Instanton). A **candidate audit instanton** is a triple (γ, b, d) , where:

- (i) $\gamma \subset \mathcal{M}$ is a closed 1-cycle representing a class $[\gamma] \in H_1(\mathcal{M}_\varepsilon; \mathbb{Z})$;

- (ii) (b, d) is the persistence interval associated to $[\gamma]$, with birth b and death d ;
- (iii) $d > \rho_{\text{crit}}$ (the cycle persists beyond the critical density threshold), indicating the cycle is **not a noise artifact**.

The candidate is **charged** if additionally $\oint_{\gamma} A > \eta$, i.e., the audit 1-form has non-trivial holonomy along γ .

define 4.2 Instanton (γ, b, d) where (i) $\gamma \subset \mathcal{M}$ 1- $H_1(\mathcal{M}_{\varepsilon}; \mathbb{Z})$ (ii) $(b, d) [\gamma]$ (iii) $d > \rho_{\text{crit}}$ $\oint_{\gamma} A > \eta$

Proposition 4.3 (Filtration-Persistence Correspondence / -). *If γ is an audit instanton in the sense of Definition 3.5, then γ corresponds to a 1-cycle in the density filtration with:*

$$\text{Birth}(\gamma) \leq \rho_{\text{crit}} \quad \text{and} \quad \text{Death}(\gamma) \geq \max_{x \in \text{interior}(\gamma)} \rho(x), \quad (21)$$

where $\text{interior}(\gamma)$ denotes the minimal spanning surface (2-chain) bounded by γ in $\mathcal{M}$.

In particular, the persistence satisfies $(\gamma) \geq \max_{x \in \text{interior}(\gamma)} \rho(x) - \rho_{\text{crit}}$.

Proposition 4.3- γ define 3.5 Instanton γ 1- $\leq \rho_{\text{crit}} \geq \max_{x \in \text{interior}(\gamma)} \rho(x)$ in particular satisfying $(\gamma) \geq \max_{x \in \text{interior}(\gamma)} \rho(x) - \rho_{\text{crit}}$

4.3 The Main Structural Theorem

theorem

We now state and prove the central result of this paper.

Theorem 4.4 (Existence of Audit Instanton Candidates / Instanton existence). *Let $(\mathcal{M}, \rho,)$ be a Situs manifold with the following properties:*

- (H1) **Smoothness** / : $\rho \in C^2(\mathcal{M}, \mathbb{R}_{\geq 0})$ and $\mathcal{M}$ has non-empty 1-homology: $\text{rank } H_1(\mathcal{M}; \mathbb{Z}) \geq 1$.
- (H2) **Bounded Expert Deviation** / : There exists a constant $K > 0$ such that $\|\nabla \bar{f}\|_{L^\infty(\mathcal{M})} < K$, where $\bar{f}$ is the mean signed deviation.
- (H3) **Non-Degeneracy** / : The critical density satisfies $\rho_{\text{crit}} > 0$ and the sublevel set $\mathcal{M}_{\rho_{\text{crit}}}$ is non-empty and has non-trivial 1-homology (at least one “hole” survives).
- (H4) **Sufficient Expert Pool** / : $M \geq 2$, and expert deviations are not perfectly anti-correlated (to exclude the trivial case where all deviations cancel exactly).

Then every non-trivial 1-homology class $0 \neq [\alpha] \in H_1(\mathcal{M}_{\rho_{\text{crit}}}; \mathbb{Z})$ contains at least one **candidate audit instanton**: there exists a 1-cycle γ representing $[\alpha]$ such that (γ, b, d) is a candidate in the sense of Definition 4.2.

Moreover, if the expert deviation field satisfies the positivity condition:

$$\forall i \in \{1, \dots, M\}, \forall x \in \mathcal{M}_{\rho_{\text{crit}}} : f_i(x) \cdot \bar{f}(x) \geq 0, \quad (22)$$

(i.e., all experts have the same sign of deviation on low-density regions), then the candidate is **charged**: $\oint_{\gamma} A > \eta$ for some $\eta > 0$ depending only on ρ_{crit} and K .

theorem 4.4 Instanton existence $(\mathcal{M}, \rho,)$ satisfying Situs (H1) $\rho \in C^2$ $\text{rank } H_1(\mathcal{M}; \mathbb{Z}) \geq 1$ (H2) $\|\nabla \bar{f}\|_{\infty} < K$ (H3) $\rho_{\text{crit}} > 0$ $\mathcal{M}_{\rho_{\text{crit}}}$ 1- (H4) $M \geq 2$ $\mathcal{M}_{\rho_{\text{crit}}}$ each 1-Instanton furthermore satisfying all

Proof / Proof. We proceed in three steps.

Step 1. Representation of the homology class. Let $0 \neq [\alpha] \in H_1(\mathcal{M}_{\rho_{\text{crit}}}; \mathbb{Z})$. By standard results in differential topology [9], there exists a smooth embedded closed curve $\gamma_0 : S^1 \rightarrow \mathcal{M}_{\rho_{\text{crit}}}$ representing $[\alpha]$. Since $\mathcal{M}_{\rho_{\text{crit}}}$ is an open submanifold of $\mathcal{M}$ (as ρ is continuous), we may perturb γ_0 slightly to ensure transversality with respect to the foliation defined by the level sets of ρ .

Step 2. Persistence in the density filtration. Consider the image of $[\alpha]$ under the inclusion $\iota_\varepsilon : \mathcal{M}_{\rho_{\text{crit}}} \hookrightarrow \mathcal{M}_\varepsilon$ for $\varepsilon \geq \rho_{\text{crit}}$. The class $[\alpha]$ is born at some density $b \leq \rho_{\text{crit}}$ (by definition, it exists in $\mathcal{M}_{\rho_{\text{crit}}}$) and dies at some $d \geq \rho_{\text{crit}}$ when it becomes a boundary. Since $\text{rank } H_1(\mathcal{M}_\varepsilon; \mathbb{Z})$ is non-decreasing in ε (by functoriality of homology) and $[\alpha]$ is non-trivial in $\mathcal{M}_{\rho_{\text{crit}}}$, its death must satisfy $d > \rho_{\text{crit}}$. Thus γ_0 corresponds to a 1-cycle in the persistence barcode with $(\gamma_0) = d - b > 0$, satisfying condition (iii) of Definition 4.2.

Step 3. Charging condition. We now verify the holonomy condition. Under hypothesis (H2), the audit 1-form $A = d_0 \bar{f}$ satisfies $\|A\|_{L^\infty(\mathcal{M})} < K$. For any 1-cycle γ ,

$$\left| \oint_\gamma A \right| \leq K \cdot (\gamma). \quad (23)$$

Under the positivity condition (22), for any x in $\mathcal{M}_{\rho_{\text{crit}}}$, all $f_i(x)$ have the same sign as $\bar{f}(x)$. This implies that on $\mathcal{M}_{\rho_{\text{crit}}}$, the absolute mean deviation $|\bar{f}|$ is bounded below by a positive constant $\delta = \min_{x \in \mathcal{M}_{\rho_{\text{crit}}}} |\bar{f}(x)| > 0$ (using compactness of $\mathcal{M}$ and continuity of $\bar{f}$).

Now, consider a refined representative γ of $[\alpha]$ that minimizes the length-to-holonomy ratio among all representatives. By the isoperimetric inequality on $\mathcal{M}$, there exists a constant $C_{\mathcal{M}} > 0$ such that for any non-trivial 1-cycle γ representing $[\alpha]$:

$$\left| \oint_\gamma A \right| \geq \frac{\delta \cdot \inf_{\Sigma: \partial \Sigma = \gamma} \text{vol}(\Sigma)}{C_{\mathcal{M}} \cdot (\gamma)}, \quad (24)$$

where Σ is any 2-chain bounding γ in $\mathcal{M}$.

Since $[\alpha] \neq 0$ in $H_1(\mathcal{M}_{\rho_{\text{crit}}}; \mathbb{Z})$, any surface Σ bounding γ must intersect the complement $\mathcal{M} \setminus \mathcal{M}_{\rho_{\text{crit}}}$ (i.e., high-density regions), and therefore has volume at least $\text{vol}(\mathcal{M}_{\rho_{\text{crit}}}) > 0$. Hence:

$$\left| \oint_\gamma A \right| \geq \eta := \frac{\delta \cdot \text{vol}(\mathcal{M}_{\rho_{\text{crit}}})}{C_{\mathcal{M}} \cdot \text{diam}(\mathcal{M})} > 0. \quad (25)$$

This establishes condition (AI3) of Definition 3.5 with the explicit threshold η given above.

Finally, condition (AI2) (local exactness) follows from Proposition 3.4 and the smoothness of $\bar{f}$ on the compact set γ , which guarantees bounded variation of the local primitive.

define 4.2 satisfying theorem all We have thus constructed, for every non-trivial $[\alpha] \in H_1(\mathcal{M}_{\rho_{\text{crit}}}; \mathbb{Z})$, a candidate audit instanton γ representing $[\alpha]$, completing the proof. $\square$

Corollary 4.5 (Minimal Number of Audit Instantons / Instanton). *Under hypotheses (H1)–(H4), the minimum number of distinct audit instanton candidates in $(\mathcal{M}, \rho,)$ is at least $\text{rank } H_1(\mathcal{M}_{\rho_{\text{crit}}}; \mathbb{Z})$ — the first Betti number of the low-density region.*

corollary 4.5 Instanton assumption (H1)–(H4) $(\mathcal{M}, \rho,)$ Instanton $\text{rank } H_1(\mathcal{M}_{\rho_{\text{crit}}}; \mathbb{Z})$ — Betti

5 Invisibility to Yajie Consensus Scoring

Yajie

The Yajie consensus score $\text{Yajie}(x)$ at a point $x \in \mathcal{M}$ is a local functional of the expert outputs $\{E_i(x)\}_{i=1}^M$ that measures the degree of agreement among experts. It is typically defined as (the negative of) the variance, or via the discrete Hodge norm of the deviation field [1]:

$$\text{Yajie}(x) = 1 - \frac{\text{Var}_i(f_i(x))}{\text{Var}_{\max}}, \quad (26)$$

where $\text{Var}_{\max}$ is a normalization constant.

YajieScore Yajie(x) Consistencydefine

Theorem 5.1 (Yajie Blindness to Audit Instantons / YajieInstanton). *Let γ be a charged audit instanton satisfying Definition 3.5 with positivity condition (22). Then:*

(i) **Pointwise consensus is high** / : *There exists $\theta > 0$ such that for all $x \in \gamma$:*

$$\text{Yajie}(x) \geq 1 - \frac{K^2 \cdot (\text{diam}(U_x))^2}{M \cdot \text{Var}_{\max}} > 1 - \theta, \quad (27)$$

where U_x is the local neighborhood of x on which A is exact.

(ii) **Cycle-integrated error is high** / : *The integrated deviation around γ is bounded below:*

$$\frac{1}{|\gamma|} \oint_{\gamma} |\bar{f}(x)| dx \geq \frac{|\Phi(\gamma)|}{|\gamma|} > \frac{\eta}{|\gamma|}, \quad (28)$$

where $\Phi(\gamma) = \oint_{\gamma} A$ is the holonomy.

(iii) **Ratio diverges with persistence** / : *As $(\gamma) \rightarrow \infty$, the ratio*

$$\frac{\text{Pointwise Yajie estimate of audit quality}}{\text{Cycle-integrated ground-truth audit quality}} \rightarrow \infty. \quad (29)$$

theorem 5.1 YajieInstanton γ satisfyingdefine 3.5 Instanton(i) Yajie γ Score(ii) γ lower bound(iii) Growth

Proof. For (i): By (AI2), on each local neighborhood U_x , $A = d_0 g_x$ with $\|\nabla g_x\|_{L^\infty} \leq C$. Hence the variation of $\bar{f}$ across U_x is bounded by $C \cdot \text{diam}(U_x)$. The variance of $\{f_i\}$ on U_x is then bounded by $O(\text{diam}(U_x)^2)$, since all experts deviate similarly on the low-density region (positivity condition). The Yajie score approaches 1 as $\text{diam}(U_x) \rightarrow 0$ or as $M \rightarrow \infty$.

For (ii): By the holonomy condition (AI3), $\oint_{\gamma} A = \oint_{\gamma} d_0 \bar{f}$ is non-zero. Since $\bar{f}$ is continuous on the compact curve γ , there exists a lower bound on $|\bar{f}|$ proportional to $|\Phi(\gamma)|/|\gamma|$.

For (iii): As $(\gamma) \rightarrow \infty$, the cycle γ wraps around a region where ρ remains arbitrarily small over a large range. In this regime, the local Yajie score is maximized (all experts agree due to shared extrapolation bias), while the integrated error is minimized only down to the topological lower bound — achieving a finite positive floor. The ratio thus diverges. $\square$

Proof(i) (AI2)each U_x $A = d_0 g_x$ $\|\nabla g_x\|_{\infty} \leq C$ $\bar{f}$ all YajieScore1(ii) (AI3) $\oint_{\gamma} A \neq 0$ γ $|\bar{f}|$ lower bound(iii) $(\gamma) \rightarrow \infty$ γ ρ any Yajielower bound

Remark 5.2 (The Non-Perturbative Analogy, Revisited /). Theorem 5.1 is the precise analog of the statement that instanton contributions $\sim \exp(-1/g^2)$ are invisible to perturbation theory: the pointwise Yajie audit (the “perturbative expansion”) assigns high scores because each local patch looks consistent, yet the global topological charge $\Phi(\gamma)$ (the “instanton number”) reveals systematic error.

The key difference from a generic correlated failure is the *topological robustness*: an audit instanton cannot be “audited away” by adding more data points in the vicinity — it requires filling the topological hole, i.e., adding data points *inside* the cycle, which may be geometrically or physically impossible in many SCX applications.

remarktheorem 5.1 Instanton $\sim \exp(-1/g^2)$ robustnessInstanton“—i.e. SCX

6 Detection Protocol

We now provide a concrete, algorithmically implementable protocol for detecting audit instantons in real SCX deployments. The protocol assumes a discretized Situs manifold represented as a simplicial complex K with N vertices (data points).

SCXInstantonassumptionSitus N K

Protocol 6.1 (Audit Instanton Detection (AID) / Instanton). **Input** / :

- Simplicial complex K approximating $\mathcal{M}$, with vertex set $V = \{v_1, \dots, v_N\}$;
- Estimated data density $\hat{\rho} : V \rightarrow \mathbb{R}_{\geq 0}$ (e.g., kernel density estimate);
- Expert outputs $\{E_i(v)\}_{i=1}^M$ for all $v \in V$;
- Critical density threshold ρ_{crit} ;
- Persistence significance threshold $\tau > 0$;
- Holonomy (circulation) threshold $\eta > 0$.

6.1Instanton $K \mathcal{M} \hat{\rho} \{E_i\} \rho_{\text{crit}} \tau \eta$

Output / : A set $\mathcal{I}$ of detected audit instanton candidates, each specified as a 1-cycle with associated metadata (persistence, holonomy, consensus score, cycle geometry).

Instanton $\mathcal{I}$ each1-Score

Step 1: Density Estimation and Filtration / 1

1. Estimate $\hat{\rho} : V \rightarrow \mathbb{R}_{\geq 0}$ using kernel density estimation (KDE) with bandwidth h chosen by cross-validation:

$$\hat{\rho}(v) = \frac{1}{Nh^n} \sum_{j=1}^N K\left(\frac{\|v - v_j\|}{h}\right). \quad (30)$$

2. Extend $\hat{\rho}$ to K via linear interpolation on simplices.

3. Construct the **density sublevel filtration**: for each vertex v , assign filtration value $\hat{\rho}(v)$; for each edge/simplex, assign the maximum density of its vertices (lower-star filtration).

each v $\hat{\rho}(v)/$

Step 2: Persistent Homology Computation / 2

3. Compute the 1-dimensional persistent homology $\text{PH}_1(K, \hat{\rho})$ using the standard matrix reduction algorithm [7] or Ripser [10].

Ripser1 $\text{PH}_1(K, \hat{\rho})$

4. Extract the persistence diagram $\mathcal{D}_1 = \{(b_j, d_j)\}_j$ and identify **significant 1-cycles**: those with persistence $d_j - b_j > \tau$.

$\mathcal{D}_1$ **1-** $d_j - b_j > \tau$

5. For each significant 1-cycle, compute a **geometric representative** γ_j — a closed edge-path in K that represents the homology class. This can be done via the “representative cycles” functionality of standard PH libraries.

each γ_j — K

Step 3: Audit 1-Form and Holonomy / 31-

6. Compute the mean signed deviation $\bar{f} : V \rightarrow \mathbb{R}$:

$$\bar{f}(v) = \frac{1}{M} \sum_{i=1}^M f_i(v), \quad f_i(v) = \langle E_i(v) - y(v), \mathbf{n}(v) \rangle. \quad (31)$$

$$\bar{f}(v) = M^{-1} \sum_i f_i(v)$$

7. Define the **discrete audit 1-form** A on each oriented edge $e = [u, v]$:

$$A(e) = \bar{f}(v) - \bar{f}(u). \quad (32)$$

This is precisely $(d_0 \bar{f})(e)$ in discrete Hodge theory.

define 1- $A(e) = \bar{f}(v) - \bar{f}(u)$ *Hodge* $(d_0 \bar{f})(e)$

8. For each candidate cycle $\gamma_j = (v_0, v_1, \dots, v_k = v_0)$, compute the **holonomy** (discrete circulation):

$$\Phi(\gamma_j) = \sum_{\ell=0}^{k-1} A([v_\ell, v_{\ell+1}]). \quad (33)$$

$$\text{each } \gamma_j \Phi(\gamma_j) = \sum_{\ell} A([v_\ell, v_{\ell+1}])$$

Step 4: Consensus Score Verification / 4Score

9. For each cycle γ_j with $|\Phi(\gamma_j)| > \eta$, compute the mean and minimum pointwise Yajie score along γ_j :

$$\text{Yajie}(\gamma_j) = \frac{1}{|\gamma_j|} \sum_{v \in \gamma_j} \text{Yajie}(v), \quad \text{Yajie}_{\min}(\gamma_j) = \min_{v \in \gamma_j} \text{Yajie}(v). \quad (34)$$

satisfying $|\Phi(\gamma_j)| > \eta$ *each* γ_j YajieScore

10. Classify cycle γ_j as a **charged audit instanton** if:

$$\text{Yajie}_{\min}(\gamma_j) > \text{Yajie}_{\text{threshold}} \quad \text{and} \quad |\Phi(\gamma_j)| > \eta. \quad (35)$$

This is the signature: high local consensus PLUS non-trivial global holonomy.

$$\text{Yajie}_{\min}(\gamma_j) > \text{Yajie}_{\text{threshold}} \quad |\Phi(\gamma_j)| > \eta \quad \gamma_j \text{ *Instanton*}$$

Step 5: Independent Verification / 5

11. For each charged audit instanton γ_j , flag the region “interior” of γ_j (the minimal spanning surface Σ_j bounded by γ_j) for **independent verification**: additional ground-truth labeling, third-party expert review, or higher-fidelity computational methods.

$$\text{eachInstanton } \gamma_j \quad \gamma_j \text{ “”} \gamma_j \quad \Sigma_j$$

12. Report all detected audit instantons with metadata: persistence (b, d) , holonomy Φ , Yajie scores, cycle geometry, and recommended verification priority (ordered by $|\Phi|$).

$$\text{allInstanton } (b, d) \quad \Phi \text{ YajieScore } |\Phi|.$$

6.1 Computational Complexity

The dominant cost is persistent homology computation, which runs in $O(|K|^\omega)$ time for a simplicial complex K , where $\omega \approx 2.376$ is the matrix multiplication exponent. In practice, for complexes arising from $N \sim 10^3$ – 10^4 data points, computation takes seconds to minutes on standard hardware using optimized libraries (GUDHI [11], Ripser [10]). The remaining steps (1-form computation, holonomy, Yajie scores) are $O(N + |E|)$ where $|E|$ is the number of edges.

$$O(|K|^\omega) \text{ for } N \sim 10^3\text{--}10^4 \quad \text{GUDHI Ripser 1-YajieScore } O(N + |E|)$$

7 Examples in SCX Auditing Scenarios

SCX

7.1 Example 1: MLIP for AlN/GaN Materials

1AlN/GaNMLIP

Consider the SCX audit of a machine-learned interatomic potential (MLIP) for AlN, as described in [1]. The Situs manifold $\mathcal{M} \subset \mathbb{R}^3$ is parametrized by (bond angle, bond length, coordination number). Three experts are audited:

- Expert A: trained only on AlN-NaCl phase space;
- Expert G: trained only on GaN-wurtzite phase space;
- Expert E: trained on full AlGaN phase space.

In the region $\Sigma = \{(\theta, r, \kappa) : \theta \in [115^\circ, 125^\circ], r \in [1.75, 1.85] \text{ \AA}\}$ (corresponding to hexagonal graphitic coordination), **no expert has training data**. All three experts extrapolate and, due to shared model-class inductive biases, arrive at similar (wrong) predictions.

AlNMLIPSCX Situs $\mathcal{M} \subset \mathbb{R}^3$ *AlN-NaClPhase Space* *GGaN-Phase Space* *AlGaNPhase Space* $\Sigma = \{(\theta, r, \kappa) : \theta \in [115^\circ, 125^\circ], r \in [1.75, 1.85] \text{ \AA}\}$ all

Example 7.1 (The Hexagonal Coordination Instanton / Instanton). The data density ρ on Σ satisfies $\rho|_\Sigma < 10^{-3}$ (compared to $\sim 10^{-1}$ in well-sampled regions). The sublevel set $\mathcal{M}_{10^{-3}}$ contains Σ and forms a topological annulus (1-hole) in the (angle, length) projection — a non-contractible 1-cycle γ encircling the hexagonal region.

Persistence homology reveals a 1-cycle with $\gamma = 0.05$ (birth at $\varepsilon = 10^{-3}$, death at $\varepsilon = 0.05$), significantly above the noise floor. The holonomy $\Phi(\gamma) = 0.47$ indicates systematic deviation around the cycle.

The pointwise Yajie score on γ averages 0.89 (high consensus!), yet PBE-DFT ground truth shows mean absolute error of 0.31 eV/atom — a clear audit instanton.

7.1 Instanton Σ $\rho < 10^{-3}$ $\mathcal{M}_{10^{-3}}$ Σ — 1- $\gamma = 0.05$ 10^{-3} 0.05 1- $\Phi(\gamma) = 0.47$ γ YajieScore 0.89 PBE-DFT 0.31 eV/atom — Instanton

7.2 Example 2: Adversarial Correlated Failure

2

Consider an SCX audit where two experts (E_1, E_2) have been adversarially manipulated to agree on a specific data submanifold Σ known to be underrepresented in training. A third honest expert E_3 disagrees but is outvoted by the Yajie majority rule.

The 1-cycle γ bounding Σ is an audit instanton: E_1 and E_2 agree (local consistency, AI2), their combined error circulates non-trivially around γ (global non-triviality, AI3), and the region is data-sparse (AI1). Persistent homology detects the hole, and the holonomy computation reveals the anomaly even though Yajie assigns high consensus based on the 2-to-1 majority.

This illustrates the **robustness** of audit instanton detection: it works even when the majority of experts are systematically wrong — precisely the scenario where standard consensus-based audit fails most catastrophically.

$E_1, E_2 \Sigma$ SCX E_3 Yajie Σ 1- γ Instanton $E_1 E_2$ Consistency γ — Yajie 21

7.3 Example 3: The SCX “Information Trap”

3 SCX “”

A recurring phenomenon in SCX auditing is the **information trap** [1]: a region of the Situs manifold where all experts agree (high Yajie score), no ground truth is available, and yet the predictions are systematically wrong. Audit instantons provide a topological characterization of information traps: an information trap is precisely the interior of a charged audit instanton cycle.

Proposition 7.2 (Information Trap = Audit Instanton Interior / = Instanton). *A connected component $\Omega \subset \mathcal{M}$ is an **information trap** in the sense of SCX if and only if its boundary $\partial\Omega$ contains a charged audit instanton γ :*

$$\oint_\gamma A > \eta, \quad \inf_{x \in \gamma} \text{Yajie}(x) > \text{Yajie}_{\text{threshold}}. \quad (36)$$

Proposition 7.2 = Instanton SCX $\Omega \subset \mathcal{M}$ if and only if $\partial\Omega$ Instanton γ

8 Relation to Discrete Hodge Theory

Hodge

Audit instantons are naturally situated within the discrete Hodge framework of SCX. We establish the precise relationship here.

8.1 The Hodge Decomposition of the Audit 1-Form

1-Hodge

Recall the Hodge decomposition on a compact Riemannian manifold:

$$\Omega^1(\mathcal{M}) = \text{im}(d_0) \oplus \mathcal{H}^1(\mathcal{M}) \oplus \text{im}(\delta_2), \quad (37)$$

where $\mathcal{H}^1(\mathcal{M}) = \ker(\Delta_1) \cong H_{\text{dR}}^1(\mathcal{M})$ is the space of harmonic 1-forms. For the audit 1-form $A = d_0 \bar{f}$, we clearly have $A \in \text{im}(d_0)$, so the harmonic component vanishes:

$$A = A_{\text{exact}} + A_{\text{harmonic}} + A_{\text{coexact}} = d_0 \bar{f} + 0 + 0. \quad (38)$$

RiemannHodge $\Omega^1(\mathcal{M}) = \text{im}(d_0) \oplus \mathcal{H}^1 \oplus \text{im}(\delta_2)$ where $\mathcal{H}^1 \cong H_{\text{dR}}^1$ 1-1- $A = d_0 \bar{f}$ $A \in \text{im}(d_0)$

Proposition 8.1 (Cohomological Signature of Audit Instantons / Instanton). *An audit instanton γ manifests in discrete Hodge theory as follows: the cycle γ is a **1-cycle with non-zero evaluation** on the harmonic representative of the cohomology class dual to $[\gamma]$. More precisely, let $\alpha \in \mathcal{H}^1(\mathcal{M})$ be the harmonic 1-form Poincaré-dual to $[\gamma]$. Then:*

$$\oint_{\gamma} \alpha = \int_{\mathcal{M}} \alpha \wedge \star \eta_{\gamma} \neq 0, \quad (39)$$

where η_{γ} is the Thom form of the normal bundle of γ .

However, since $A = d_0 \bar{f}$ is exact, its pairing with α is zero: $\oint_{\gamma} A = 0$ would hold if $\bar{f}$ were globally defined. The non-zero holonomy signals that $\bar{f}$ is not globally single-valued — it has monodromy around the 1-cycle γ .

Proposition 8.1 *InstantonHodgeInstanton γ $[\gamma]$ 1-howeverbecause $A = d_0 \bar{f}$ $\bar{f}$ —1- γ*

8.2 Discrete Laplacian and Audit Instanton Spectrum

LaplacianInstanton

Proposition 8.2 (Spectral Detection /). *Let L_1 be the discrete 1-Laplacian on the simplicial complex K approximating $\mathcal{M}$. The non-trivial 1-cycles corresponding to audit instanton candidates lie in $\ker(L_1)$, the space of harmonic 1-chains:*

$$\ker(L_1) = \ker(d_1) \cap \ker(\delta_1) \cong H_1(K; \mathbb{R}). \quad (40)$$

Moreover, if A is the discrete audit 1-cochain, then the projection of A onto $\ker(L_1)$ is non-zero precisely when there exists at least one audit instanton candidate in K :

$$\|P_{\ker(L_1)} A\| > 0 \iff \exists \text{ at least one charged audit instanton candidate.} \quad (41)$$

Proposition 8.2 *$L_1 \mathcal{M} K$ 1-LaplacianInstanton1- $\ker(L_1)$ 1-furthermore $A \in \ker(L_1)$ if and only if K there existsInstanton*

9 Discussion and Open Problems

Open Problem

9.1 Summary /

We have introduced **audit instantons** as a rigorous mathematical framework for detecting non-perturbative, topologically protected expert consensus failures in the SCX auditing paradigm. The central insight — that pointwise audit is structurally analogous to perturbative expansion, and thus systematically misses globally correlated topological defects — leads to a concrete detection protocol using persistent homology, with provable guarantees.

The key contributions are:

- (1) A cohomological definition of audit instantons (Definition 3.5);
- (2) A structural theorem linking homology classes to audit instanton candidates (Theorem 4.4);
- (3) A proof of invisibility to pointwise Yajie consensus (Theorem 5.1);
- (4) An implementable detection protocol (Protocol 6.1).

*we introduce **InstantonSCXCore** — — therefore Missing — — Proof*

9.2 Relation to the String Theory Analogy

It is important to clarify the *nature* of the analogy with string theory worldsheet instantons. We do **not** claim a formal mathematical correspondence between audit instantons and worldsheet instantons. Rather, the analogy operates at the level of **structural principles**:

Concept	String Theory	SCX Audit <i>SCX</i>
Non-perturbative effect	Worldsheet instanton <i>Instanton</i>	Audit instanton <i>Instanton</i>
Perturbative expansion	Genus expansion	Pointwise Yajie audit <i>Yajie</i>
Suppression factor	$\exp(-A/\alpha')$ $\exp(-A/\alpha')$	$\exp(-C/\sigma^2)$ $\exp(-C/\sigma^2)$
Topological classification	π_n or H_n π_n H_n	H_1 (persistent) H_1
Instanton number <i>Instanton</i>	Winding / Chern class / <i>Chern</i>	Holonomy $\Phi(\gamma)$ $\Phi(\gamma)$
Detection	Moduli space geometry	Persistent homology

The analogy is **heuristic in origin but rigorous in execution**: it motivated the search for a topological characterization of non-local audit failures, and the resulting mathematical framework stands independently of string theory.

*InstantonInstantonInstantonthere exists **Heuristic***

9.3 Open Problems

Open Problem

We identify several open problems for future investigation:

- OP1 Sharpness of bounds / :** Can the lower bound η in Theorem 4.4 be sharpened using more refined geometric data (curvature of $\mathcal{M}$, higher-order statistics of expert deviations)?
- OP2 Higher-dimensional audit instantons / Instanton:** Can the definition be extended to k -cycles for $k \geq 2$? A “2-audit instanton” would correspond to a 2-cycle (void/bubble) in the density filtration, representing a region where expert consensus *encloses* a systematically wrong prediction volume.
- OP3 Dynamical audit instantons / Instanton:** In the time-dependent SCX framework (with evolving Spring gatekeeper and drifting experts), do audit instantons exhibit dynamics — creation, annihilation, merging? What is the analog of the “instanton gas” approximation?
- OP4 Statistical significance / :** Develop hypothesis tests for the null hypothesis “the holonomy is consistent with zero” vs. the alternative “there exists an audit instanton,” accounting for the adaptive nature of the cycle selection.
- OP5 Algorithmic efficiency / :** Can the detection be accelerated by first computing the 1-Laplacian eigenmodes and restricting persistence computation to the subspace of near-harmonic cycles (the “topological subspace”)?
- OP6 Empirical validation / :** Apply Protocol 6.1 to real SCX audit datasets (AIN MLIP, PDF reconstruction, ACE energy surfaces) and measure the prevalence and impact of audit instantons compared to isolated pointwise expert disagreements.
- OP7 Connection to adversarial robustness / robustness:** Are audit instantons related to adversarial examples in deep learning? Both involve low-data-density regions and systematic model failures. Can audit instanton detection be repurposed as an adversarial robustness diagnostic?
- OP8 Connection to Morse theory / Morse:** If ρ is treated as a Morse function on $\mathcal{M}$, the birth and death of 1-cycles in the persistence diagram correspond to critical points of ρ . Can the “instanton action” $\sim \exp(-1/\rho)$ be given a Morse-theoretic interpretation?

we point out that Open Problem(OP1) (OP2) Instanton $k \geq 2$ (OP3) Instanton(OP4) (OP5) (OP6) SCX(OP7) robustness(OP8) Morse

10 Conclusion

Conclusion

Conclusion

Conclusion

We have introduced the concept of **audit instantons** — non-perturbative, topologically protected failures of expert consensus — and provided a rigorous mathematical framework for their definition, detection, and interpretation within the SCX auditing paradigm.

The framework builds on persistent homology of the data density filtration and discrete Hodge theory, with structural inspiration from worldsheet instantons in string theory.

Audit instantons reveal a fundamental limitation of pointwise consensus auditing: by treating each data point independently, it systematically misses globally correlated failure modes concentrated in low-density regions of the data manifold. These failures are *topologically robust* — they cannot be eliminated by adding more auditors or adjusting consensus thresholds, but require filling the topological “holes” in the data distribution.

The detection protocol we provide is concrete, algorithmically implementable, and computationally tractable for realistic SCX deployments. We believe audit instantons open a new direction in trustworthy AI auditing — one that moves beyond local perturbation theory to a genuinely topological understanding of when and why expert systems fail.

*we introduce Instanton — — — SCX define Hodge Instanton
Instanton each Missing — — — “”
SCX Instanton AI — — Why*

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SCX Xiaogan SCX Instanton “SCX” GUDHI Ripser

A Proof of the Holonomy Lower Bound

lower bound Proof

We provide a more detailed proof of the isoperimetric inequality used in Theorem 4.4, showing that $\oint_{\gamma} A > \eta$ under the positivity condition.

Lemma A.1 (Isoperimetric Inequality on $\mathcal{M} / \mathcal{M}$ awaiting awaiting). *Let $\mathcal{M}$ be a compact Riemannian n -manifold with Ricci curvature bounded below by $-(n-1)\kappa^2$ ($\kappa \geq 0$). For any 1-cycle γ representing a non-trivial homology class $[\gamma] \in H_1(\mathcal{M}; \mathbb{Z})$, there exists a constant $C_{\mathcal{M}} > 0$ depending only on the geometry of $\mathcal{M}$ such that for any 2-chain Σ with $\partial\Sigma = \gamma$:*

$$\text{vol}_{n-1}(\Sigma) \geq C_{\mathcal{M}} \cdot \frac{\left| \oint_{\gamma} A \right|}{\|A\|_{L^\infty} \cdot (\gamma)} \cdot \text{vol}_n(\text{interior supported region}). \quad (42)$$

Proof. By the coarea formula and the fact that $A = d_0 \bar{f}$, we have:

$$\oint_{\gamma} A = \int_{\gamma} d_0 \bar{f} = \int_{\Sigma} d_1 d_0 \bar{f} = 0, \quad (43)$$

if $\Sigma \subset \mathcal{M}$. This apparent contradiction is resolved by noting that *no* 2-chain Σ in $\mathcal{M}_{\rho_{\text{crit}}}$ bounds γ — indeed, condition (AI3) asserts precisely that γ is not a boundary in the low-density region.

For any 2-chain $\Sigma \subset \mathcal{M}$ with $\partial\Sigma = \gamma$, Σ must intersect $\mathcal{M} \setminus \mathcal{M}_{\rho_{\text{crit}}}$ (the high-density region). Applying the isoperimetric inequality on $\mathcal{M}$ [12], the volume of Σ is bounded

below by the **filling area** of γ , which for non-trivial homology classes scales with the injectivity radius of $\mathcal{M}$ and the homological systole.

Explicitly, let $\text{sys}_1(\mathcal{M}_{\rho_{\text{crit}}})$ denote the 1-systole of the low-density region (the minimal length of a non-trivial 1-cycle in $\mathcal{M}_{\rho_{\text{crit}}}$). Then any bounding chain Σ must have $(n-1)$ -volume at least $\text{sys}_1(\mathcal{M}_{\rho_{\text{crit}}})$, and the holonomy bound follows from the positivity condition (22), which gives a pointwise lower bound on the integrand of $\oint_{\gamma} A$. $\square$

lemma A.1 *awaiting* $\mathcal{M}$ Ricci lower bound $-(n-1)\kappa^2$ Riemann n - $[\gamma] \in H_1(\mathcal{M}; \mathbb{Z})$ 1- γ there exists $\mathcal{M}$ $C_{\mathcal{M}} > 0$ such that satisfying $\partial\Sigma = \gamma$ 2- Σ lower bound

B Discrete Formulation for Numerical Implementation

For readers implementing Protocol 6.1, we provide the discrete analogues of all key quantities.

Let K be a finite simplicial complex with:

- Vertices $V = \{v_1, \dots, v_N\}$ (audit data points);
- Edges $E = \{e_1, \dots, e_{|E|}\}$ (pairs of nearby points);
- Triangles $T = \{t_1, \dots, t_{|T|}\}$ (triples, for $n \geq 2$).

$K \ V \ E \ T$

Discrete boundary operators / :

$$(\partial_1)_{ve} = \pm 1 \text{ if } v \in \partial e \text{ (with consistent orientation),} \quad (44)$$

$$(\partial_2)_{et} = \pm 1 \text{ if } e \in \partial t. \quad (45)$$

Discrete 1-Laplacian / **1-Laplacian**:

$$L_1 = \partial_2 \partial_2^T + \partial_1^T \partial_1. \quad (46)$$

Discrete audit 1-cochain / **1-**:

$$A \in \mathbb{R}^{|E|}, \quad A(e = [u, v]) = \bar{f}(v) - \bar{f}(u). \quad (47)$$

Discrete holonomy / : For a 1-cycle $\gamma = \sum_{e \in E} c_e e$ with integer coefficients c_e , the holonomy is:

$$\Phi(\gamma) = \sum_{e \in E} c_e \cdot A(e) = c^T A. \quad (48)$$

Harmonic projection detection / : Compute the eigenvectors $\{u_j\}_{j=1}^{\dim \ker(L_1)}$ of L_1 with eigenvalue zero. Then:

$$\|P_{\ker(L_1)} A\|^2 = \sum_{j=1}^{\dim \ker(L_1)} |\langle u_j, A \rangle|^2. \quad (49)$$

A non-zero norm signals the presence of at least one audit instanton candidate.

$$L_1 \{u_j\} \|P_{\ker(L_1)} A\|^2 = \sum_j |\langle u_j, A \rangle|^2 \text{ Instanton there exists}$$

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